

Microeconomic Analysis

Multivariate Calculus

HT 2025

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Multivariate Calculus

[Pre-readings](#) on sets and univariate calculus on Canvas (notes by Prof. Keller).

There are some exercises with solutions for you to practice if you haven't done so yet.

Further readings:

- Simon, C.P. and Blume, L. *Mathematics for Economists*. Ch13-19, 30
- Pemberton, M. and Rau, N. *Mathematics for Economists*. Ch 13-17
- Sundaram, R. *A First Course in Optimization Theory*. Ch13-14
- de la Fuente, A. *Mathematical Methods and Models for Economists*. Ch1-4

Some of the content of these slides use previous material developed by Margaret Stevens, Godfrey Keller and Alan Beggs.

Multivariate Calculus

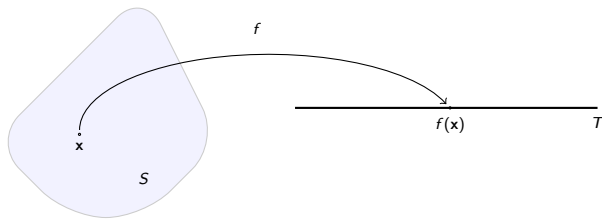
Topics

- I Functions and Continuity
- II Differential Calculus
- III The Implicit Function Theorem

I. Functions

Functions in $\mathbb{R}^n$

Given $S \subset \mathbb{R}^n$ and $T \subset \mathbb{R}$ a **real-valued function** $f : S \rightarrow T$ is a rule that associates with each element of S one and only one element of T .

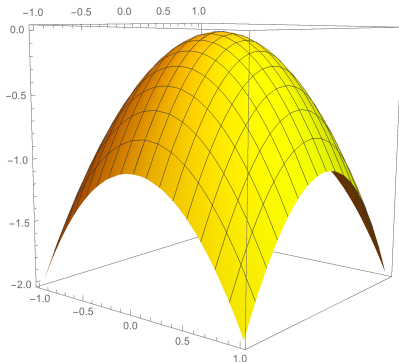


Examples:

- $f(x, y) = 2xy$ is a function from $\mathbb{R}^2$ to $\mathbb{R}$.

Functions in $\mathbb{R}^2$

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, then $\{(x, y, z) \mid z = f(x, y)\}$ describes a surface in $\mathbb{R}^3$.



$$f(x, y) = -x^2 - y^2$$

Sometimes they are represented with level curves on a plane. A **level curve** or **contour** of a function f is the set of points where f takes a given value (e.g. an indifference curve, isoquant).

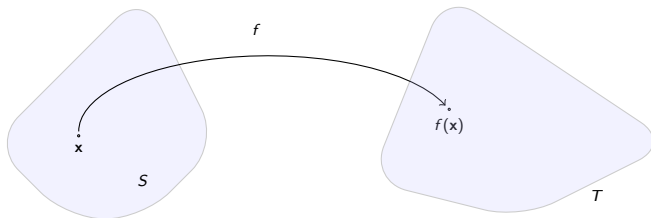
Functions in $\mathbb{R}^n$

Given $S \subset \mathbb{R}^n$ and $T \subset \mathbb{R}^m$ a **vector-valued function** $f : S \rightarrow T$ is a rule that associates with each element of S one and only one element of T .

We can write

$$f(\mathbf{x}) = \begin{pmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \\ \vdots \\ f_m(\mathbf{x}) \end{pmatrix} = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x}))^T$$

where each of the f_i are real-valued functions.

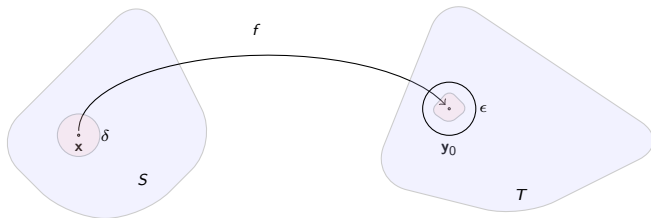


Limits of Functions in $\mathbb{R}^n$

Let $f : S \rightarrow T$, where $S \subset \mathbb{R}^n$ and $T \subset \mathbb{R}^m$, and $\mathbf{x}_0 \in S$.

Then f has a **limit**, $\mathbf{y}_0$, at $\mathbf{x}_0$ if and only if for any $\epsilon > 0$, $\exists \delta > 0$ such that

$$0 < \|\mathbf{x} - \mathbf{x}_0\| < \delta \implies \|f(\mathbf{x}) - \mathbf{y}_0\| < \epsilon \quad (\Leftrightarrow f(B_\delta(\mathbf{x}_0)) \subset B_\epsilon(\mathbf{y}_0))$$



We should be able to find δ for any ϵ .

We write: $\lim_{\mathbf{x} \rightarrow \mathbf{x}_0} f(\mathbf{x}) = \mathbf{y}_0$

Limits of Functions in $\mathbb{R}^n$

Examples:

- $f(x_1, x_2) = x_1 x_2$. $\lim_{\mathbf{x} \rightarrow \mathbf{0}} f(\mathbf{x}) = 0$:

Consider $\epsilon > 0$ and $\delta = \sqrt{\epsilon}$, for any $\mathbf{x}$ such that $\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2} < \delta$,

$$|f(\mathbf{x}) - f(\mathbf{0})| = |x_1 x_2 - 0| = |x_1 x_2| = |x_1| \cdot |x_2| < \delta^2 = \epsilon$$

Note that $|x_i| = \sqrt{x_i^2} \leq \sqrt{x_1^2 + x_2^2} = \|\mathbf{x}\|$

Limits of Functions in $\mathbb{R}^n$

Examples:

- $f(x_1, x_2) = x_1 + x_2$. $\lim_{\mathbf{x} \rightarrow (1,1)} f(\mathbf{x}) = 2$.

Consider $\epsilon > 0$, define $\delta = \frac{\epsilon}{2}$. Then, for any $\mathbf{x}$ such that

$$\|(x_1, x_2) - (1, 1)\| = \sqrt{(x_1 - 1)^2 + (x_2 - 1)^2} < \delta$$

We have that,

$$|f(x_1, x_2) - f(1, 1)| = |x_1 + x_2 - 2| = |x_1 - 1 + x_2 - 1| \leq |x_1 - 1| + |x_2 - 1| < 2\delta < \epsilon$$

$$(\text{Note that } |x_i - 1| < \sqrt{(x_1 - 1)^2 + (x_2 - 1)^2} < \delta)$$

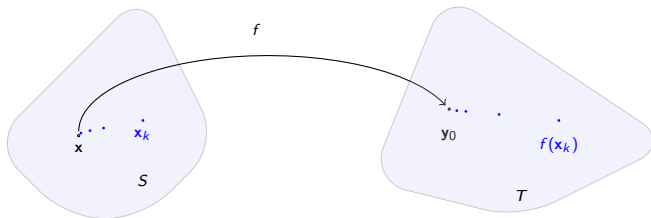
Limits of Functions in $\mathbb{R}^n$

Equivalent definition for the limit of a function using sequences:

Theorem

Let $f : S \rightarrow T$, where $S \subset \mathbb{R}^n$ and $T \subset \mathbb{R}^m$, and $\mathbf{x}_0 \in S$.

f has a **limit**, $\mathbf{y}_0$, at $\mathbf{x}_0$, if and only if for **any** sequence $\{\mathbf{x}_k\}_{k=1}^{\infty} \subset S$ converging to $\mathbf{x}_0$, the sequence $\{f(\mathbf{x}_k)\}_{k=1}^{\infty}$ converges to $\mathbf{y}_0$.



It needs to be true for all converging sequences.

Example:

$$\bullet f(x_1, x_2) = \begin{cases} \frac{x_1 x_2}{x_1^2 + x_2^2} & \text{if } (x_1, x_2) \neq (0, 0) \\ 0 & \text{if } (x_1, x_2) = (0, 0) \end{cases} . \text{ What is the } \lim_{\mathbf{x} \rightarrow (0,0)} f(\mathbf{x})?$$

- Consider the sequence $\{(0, \frac{1}{k})\}_{k=1}^{\infty}$. This sequence converges to $(0, 0)$.

$$f(0, \frac{1}{k}) = \frac{0}{\frac{1}{k^2}} = 0.$$

- Consider the sequence $\{(\frac{1}{k}, \frac{1}{k})\}_{k=1}^{\infty}$. This sequence converges to $(0, 0)$.

$$f(\frac{1}{k}, \frac{1}{k}) = \frac{\frac{1}{k^2}}{\frac{1}{k^2} + \frac{1}{k^2}} = \frac{1}{2}.$$

Hence the limit of f at $(0, 0)$ does not exist.

Limits of Functions in $\mathbb{R}^n$

The limits therefore inherit all the properties of sequences:

- The limit, if it exists, is unique.

- Given $f = (f_1, f_2, \dots, f_m)^T$,

$$\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) = \mathbf{A} \Leftrightarrow \lim_{\mathbf{x} \rightarrow \mathbf{a}} f_i(\mathbf{x}) = A_i \quad \forall i = 1, \dots, m$$

- **Algebra of Limits.** Let $f, g : S \rightarrow T$ with $S \subset \mathbb{R}^n$, and $T \subset \mathbb{R}$.

Suppose $\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) = A$ and $\lim_{\mathbf{x} \rightarrow \mathbf{a}} g(\mathbf{x}) = B$ then:

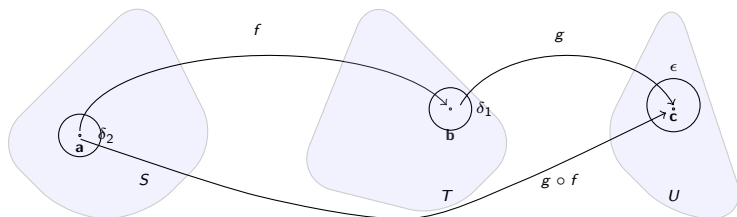
- ▶ $\lim_{\mathbf{x} \rightarrow \mathbf{a}} [f(\mathbf{x}) + g(\mathbf{x})] = A + B$
- ▶ $\lim_{\mathbf{x} \rightarrow \mathbf{a}} [f(\mathbf{x})g(\mathbf{x})] = AB$
- ▶ $\lim_{\mathbf{x} \rightarrow \mathbf{a}} [f(\mathbf{x})/g(\mathbf{x})] = A/B \quad \text{if } B \neq 0$

Limits of Functions in $\mathbb{R}^n$

Limits of Composition of functions:

- Let $f : S \rightarrow T$ and $g : T \rightarrow U$ with $S \subset \mathbb{R}^n$, $T \subset \mathbb{R}^m$ and $U \subset \mathbb{R}^k$. Suppose $\lim_{x \rightarrow a} f(x) = \mathbf{b}$ and $\lim_{y \rightarrow \mathbf{b}} g(y) = \mathbf{c}$, then

$$\lim_{x \rightarrow a} g(f(x)) = \mathbf{c}$$



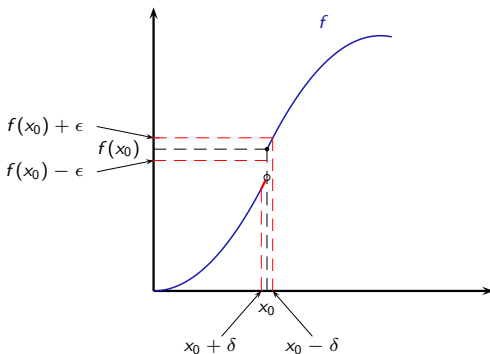
Continuity of Functions in $\mathbb{R}^n$

Let $f : S \rightarrow T$, with $S \subset \mathbb{R}^n$, and $T \subset \mathbb{R}^m$

f is **continuous** at $\mathbf{x}_0$ if and only if

$$\lim_{\mathbf{x} \rightarrow \mathbf{x}_0} f(\mathbf{x}) = f(\mathbf{x}_0)$$

If f is continuous at every point in its domain S , it is said to be continuous on S .



Continuity

Some Properties:

1. Let f and g be functions from $\mathbb{R}^n$ to $\mathbb{R}^m$. Suppose that f and g are continuous at $\mathbf{x}$. Then $f + g$, $f - g$ are all continuous at $\mathbf{x}$.
2. Let f and g be functions from $\mathbb{R}^n$ to $\mathbb{R}$. Suppose that f and g are continuous at $\mathbf{x}$, then $f \cdot g$ is continuous at $\mathbf{x}$, moreover, if $g(\mathbf{x}) \neq 0$, then f/g is continuous at $\mathbf{x}$.
3. Let $f = (f_1, \dots, f_m)^T$ be a function from $\mathbb{R}^n$ to $\mathbb{R}^m$. Then, f is continuous at $\mathbf{x}$ if and only if each of its component functions $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous at $\mathbf{x}$.
4. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $g : \mathbb{R}^m \rightarrow \mathbb{R}^k$. Suppose that f is continuous at $\mathbf{x}$ and g is continuous at $f(\mathbf{x})$. Then, the composition $g \circ f : \mathbb{R}^n \rightarrow \mathbb{R}^k$ is continuous at $\mathbf{x}$.

Continuity of Functions in $\mathbb{R}^n$

Example of application:

- $f(x) = x$ is easily seen to be continuous. Applying the product rule with $f(x) = x$ and $g(x) = x$, deduce x^2 is continuous. Applying the product rule again, so is x^3 etc.
- $g(x) = 2$ is continuous, hence applying rule the product rule with $f(x) = x$, $g(x) = 2$, $2x$ is continuous.
- Applying the addition rule, $x^2 + 2x$ is a continuous function.
- In similar fashion deduce that any polynomial function is continuous.

Some Useful Theorems for Continuous Functions

We will now present some useful theorems for real-valued continuous functions:

The Weierstrass Theorem: A continuous function $f : S \rightarrow \mathbb{R}$, where S is a compact subset of $\mathbb{R}^n$, attains a maximum and minimum value on S .

The Intermediate Value Theorem: Suppose $f : [a, b] \rightarrow \mathbb{R}$ is continuous, and $f(a) < y < f(b)$. Then $\exists c \in (a, b)$ such that $f(c) = y$.

Brouwer's Fixed Point Theorem: Suppose $f : [a, b] \rightarrow [a, b]$ is a continuous function on a closed interval in $\mathbb{R}$. Then f has a fixed point: that is $\exists x_0 \in [a, b]$ such that $f(x_0) = x_0$.

Brouwer's Fixed Point Theorem in $\mathbb{R}^n$: Suppose $S \subset \mathbb{R}^n$ is a non-empty, compact and convex set, and $f : S \rightarrow S$ is a continuous function. Then f has a fixed point: that is $\exists \mathbf{x}_0$ such that $f(\mathbf{x}_0) = \mathbf{x}_0$.

Weierstrass Theorem

Weierstrass Theorem

A **continuous** real valued function $f : S \rightarrow \mathbb{R}$, where S is a **compact** subset of $\mathbb{R}^n$ attains a **maximum** and a **minimum** value.

The Weierstrass Theorem is an **extreme value theorem**, as it says that the extreme values of the function are achieved. It is useful to show that optimal points exist.

Two main assumptions: compactness and continuity

$$f : [0, 1) \rightarrow \mathbb{R}$$

$$f(x) = x$$

S is not closed

No Max

$$f : [0, \infty) \rightarrow \mathbb{R}$$

$$f(x) = x$$

S is unbounded

No Max.

$$f : [0, 1] \rightarrow \mathbb{R}$$

$$f(x) = \begin{cases} \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

f is not continuous at 0

No Max.

Weierstrass Theorem

We will prove it for functions $f : [a, b] \rightarrow \mathbb{R}$, in two steps.

- 1 If $f : [a, b] \rightarrow \mathbb{R}$ continuous, then f is bounded, i.e., there exists M such that $|f(x)| \leq M$ for all $x \in [a, b]$.

Proof by contradiction:

- ▶ Suppose it is unbounded above, for any $k \in \mathbb{N}$, $\exists x_k \in [a, b]$ such that $f(x_k) > k$.
- ▶ $\{x_k\} \subset [a, b]$: By Bolzano-Weierstrass Theorem it has a convergent subsequence:

$$\{x_{k_j}\} \rightarrow x$$

- ▶ Since $[a, b]$ closed, $x \in [a, b]$.
- ▶ By continuity of f , $\lim_{j \rightarrow \infty} f(x_{k_j}) = f(x)$
- ▶ But $f(x_{k_j}) > k_j \rightarrow \infty$, we have reached a contradiction
- ▶ We use a similar argument if unbounded below.

□

Weierstrass Theorem

2 Now we are ready to prove the Weierstrass Theorem:

- ▶ By 1, the set $\{f(x) : x \in [a, b]\}$ is bounded above and below and hence has a supremum and an infimum
- ▶ Denote $F = \sup\{f(x) : x \in [a, b]\}$. Then for all $k \in \mathbb{N}$, there exists $x_k \in [a, b]$ such that

$$|f(x_k) - F| < \frac{1}{k}$$

- ▶ $\{x_k\} \subset [a, b]$, by Bolzano-Weierstrass it has a convergent subsequence $\{x_{k_j}\} \rightarrow x_M$
Can we show that $f(x_M) = F$?
- ▶ By continuity $\{f(x_{k_j})\} \rightarrow f(x_M)$. So for any $\epsilon > 0$, $\exists J$ such that

$$|f(x_{k_j}) - f(x_M)| < \frac{\epsilon}{2} \quad \forall j \geq J$$

- ▶ Then, given ϵ and j such that $j \geq \max\{J, \frac{2}{\epsilon}\}$,

$$|f(x_M) - F| \leq |f(x_M) - f(x_{k_j})| + |f(x_{k_j}) - F| < \frac{\epsilon}{2} + \frac{1}{k_j} < \frac{\epsilon}{2} + \frac{1}{j} < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

so $f(x_M) = F$ and the maximum is reached.

- ▶ Same argument for minimum.

□

Intermediate Value Theorem

The Intermediate Value Theorem

Suppose $f : [a, b] \rightarrow \mathbb{R}$ is continuous, and $f(a) < y < f(b)$. Then $\exists c \in (a, b)$ such that $f(c) = y$.

This result will be useful for example to prove the fix point theorems.

Proof:

- ▶ Define $c = \sup\{x \in [a, b] : f(x) \leq y\}$. c is well defined: the set is non-empty and bounded so the supremum exists
- ▶ By construction $f(x) > y$ if $x > c$, by continuity $f(c) = \lim_{x \rightarrow c+} f(x) \geq y$.
- ▶ But if $f(c) > y$, then it is not the lowest upper bound.
- ▶ Hence $f(c) = y$, and clearly $c \neq a, b$ so $c \in (a, b)$. □

Brouwer's Fixed Point Theorems

Brouwer's Fixed Point Theorems

- Suppose $f : [a, b] \rightarrow [a, b]$ is a continuous function on a closed interval in $\mathbb{R}$. Then f has a fixed point: that is $\exists x_0 \in [a, b]$ such that $f(x_0) = x_0$.
- Suppose $S \subset \mathbb{R}^n$ is a non-empty, compact and convex set, and $f : S \rightarrow S$ is a continuous function. Then f has a fixed point: that is $\exists \mathbf{x}_0$ such that $f(\mathbf{x}_0) = \mathbf{x}_0$.

These theorems are useful for proving existence of Nash equilibrium and competitive equilibrium.

Brouwer's Fixed Point Theorems

Outline the proof when $f : [0, 1] \rightarrow [0, 1]$:

Let $g(x) = x - f(x)$. Then $g(x^*) = 0$ if and only if $f(x^*) = x^*$.

- ▶ If $g(0) = 0$ then $x^* = 0$ is a fixed point of f and we are done.
If not $g(0) < 0$.
- ▶ If $g(1) = 0$ then $x^* = 1$ is a fixed point of f and we are done.
If not $g(1) > 0$.
- ▶ If neither 0 or 1 are fixed points of f , then $g(0) < 0 < g(1)$.
By the Intermediate Value Theorem, there exists $x^* \in (0, 1)$ with $g(x^*) = 0$: that is x^* is a fixed point of f .



II. Differentiation

Differentiation in $\mathbb{R}$: Recap from Pre-reading

Let $f : \mathbb{R} \rightarrow \mathbb{R}$, f is **differentiable** at x if the following limit exists:

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

If the limit exists, we call it the **derivate** of f at x , and write $f'(x)$.

Interpretation: Slope of the curve f at x .

Theorem

If f is differentiable at x , then f is continuous at x .

Proof:

Suppose $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x)$, then

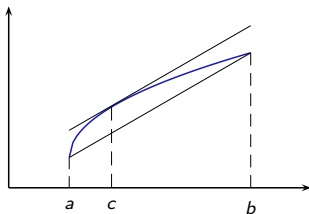
$$\begin{aligned} \lim_{h \rightarrow 0} f(x+h) - f(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \cdot h \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \cdot \lim_{h \rightarrow 0} h = f'(x) \cdot 0 = 0 \end{aligned}$$

Mean Value Theorem

The Mean Value Theorem

Suppose $f : U \rightarrow \mathbb{R}$, where U is a convex subset of $\mathbb{R}$, and f is a differentiable function. Then for any points $a, b \in U$, there is a point $c \in (a, b)$ such that

$$f(b) - f(a) = f'(c)(b - a)$$

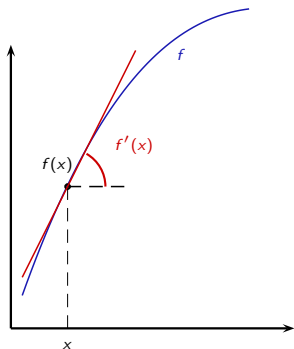


$$\underbrace{\frac{f(b) - f(a)}{b - a}}_{\text{slope of chord}} = f'(c)$$

This theorem is useful to derive some relationships between a function and its derivative. It is also used to derive the Taylor's theorems (later)

Linear Approximation in $\mathbb{R}$

If a function f is differentiable at x , it can be approximated by a linear function with slope $f'(x)$.



- If $f'(x)$ exists:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\lim_{h \rightarrow 0} \frac{f(x+h) - [f(x) + hf'(x)]}{h} = 0$$

So the tangent $f(x) + hf'(x)$ is the best linear approximation of $f(x+h)$ close to x .

Taylor Approximations in $\mathbb{R}$

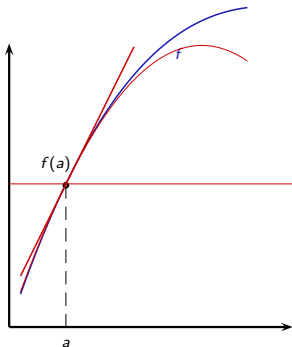
We can use higher order derivatives to improve the approximation:

0th-order: $f(x) \approx f(a)$

1st-order: $f(x) \approx f(a) + (x - a)f'(a)$

2nd-order:
$$f(x) \approx f(a) + (x - a)f'(a) + \frac{f''(a)}{2}(x - a)^2$$

k^{th} -order:
$$f(x) \approx f(a) + (x - a)f'(a) + \frac{f''(a)}{2}(x - a)^2 + \cdots + \frac{f^{(k)}(a)}{k!}(x - a)^k$$



Taylor's Theorem in $\mathbb{R}$

Taylor's Theorem for a function of one variable

Suppose $f : U \rightarrow \mathbb{R}$, where U is a convex subset of $\mathbb{R}$, is $k + 1$ times differentiable. Then for any $a, x \in U$, there exists $z \in (a, x)$ such that:

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2}(x - a)^2 + \cdots + \frac{f^{(k)}(a)}{k!}(x - a)^k + \frac{f^{(k+1)}(z)}{(k + 1)!}(x - a)^{k+1}$$

Note that the remainder $R_k(x, a) = \frac{f^{(k+1)}(z)}{(k + 1)!}(x - a)^{k+1}$ is of order $(x - a)^{k+1}$

$$R_k(x, a) = O\left((x - a)^{k+1}\right) \quad \text{or} \quad \frac{R_k(x, a)}{(x - a)^k} \rightarrow 0$$

Maclaurin's Theorem: (Special case of Taylor's Theorem with $a = 0$)

$$f(x) = f(0) + xf'(0) + \frac{f''(0)}{2}x^2 + \cdots + \frac{f^{(k)}(0)}{k!}x^k + \frac{f^{(k+1)}(z)}{(k + 1)!}x^{k+1}$$

Taylor's Series in $\mathbb{R}$

If f is infinitely differentiable, we have an infinite series. If $R_k \rightarrow 0$ as $k \rightarrow \infty$ then it converges and

$$f(x) = f(a) + (x-a)f'(a) + \frac{f''(a)}{2}(x-a)^2 + \cdots + \frac{f^{(k)}(a)}{k!}(x-a)^k + \cdots$$

Then, this is the **Taylor Expansion** of f about a .

Examples:

- $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots +$

Expand $f(x) = e^x$ around $x = 0$.

Note that $f'(x) = e^x$ so $f^{(k)}(x) = e^x$ and $f^{(k)}(0) = e^0 = 1$

Taylor's Series in $\mathbb{R}$

Examples:

- $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + \dots + (-1)^{n+1} \frac{x^n}{n} + \dots$ (valid only for $-1 < x \leq 1$)

Expand $f(x) = \ln(x+1)$ around $x=0$

Note that $f'(x) = \frac{1}{1+x}$, $f''(x) = -\frac{1}{(1+x)^2}, \dots$ $f^{(k)} = (-1)^{k+1} \frac{(k-1)!}{(1+x)^k}$

- Taylor expansion of $(1+x)^n$. If n is a positive integer we have the (finite) Binomial expansion:

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots + {}^nC_k x^k + \dots + x^n$$

For other values of n we get an infinite series, valid only for $|x| < 1$

Differentiation in $\mathbb{R}^n$

Let $f : S \rightarrow T$, where $S \subset \mathbb{R}^n$, $T \subset \mathbb{R}$ be a **real-valued** function of n variables.

The k^{th} **partial derivative** of f at $\mathbf{x} \in \mathbb{R}$ is

$$\frac{\partial f}{\partial x_k}(\mathbf{x}) = \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_k + h, \dots, x_n) - f(x_1, \dots, x_k, \dots, x_n)}{h}$$

- If the limit doesn't exist, there is no k^{th} partial derivative.
- The k^{th} partial derivative is a measure of how fast the function changes when its k^{th} argument, x_k , changes, but all the other arguments remain constant.
- The k^{th} partial derivative of f at $\mathbf{x}$ is sometimes denoted by $f_k(\mathbf{x})$, f_{x_k} , $\partial_{x_k} f$, $\partial_k f$, $D_k f$.

Gradient

Let $f : S \rightarrow T$, where $S \subset \mathbb{R}^n$, $T \subset \mathbb{R}$ be a **real-valued** function of n variables.

The vector of first derivatives, Df , is called the *Gradient* vector:

$$Df(\mathbf{x}) = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right) \quad \text{or} \quad Df(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_n(\mathbf{x}))$$

Example: If $f(x, y) = x^2y$ then $Df = (2xy, x^2)$

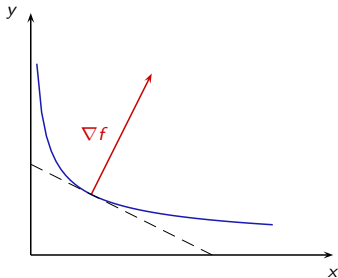
Sometimes, it is written as a column vector and denoted as ∇f :

$$\nabla f = \begin{pmatrix} \partial_1 f \\ \partial_2 f \\ \vdots \\ \partial_n f \end{pmatrix}$$

Gradient

A **level curve** (surface) or **contour** of a function f is the set of points where f takes a given value (e.g. an indifference curve, isoquant).

∇f points in the direction in which the rate of change of f is greatest and is perpendicular to the tangent plane of the level curve through a given point.



$$f(x, y) = xy^2, \nabla f = \begin{pmatrix} y^2 \\ 2xy \end{pmatrix}. \text{ At } (x, y) = (1, 1), \nabla f = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Jacobian Matrix

If $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, a **vector-valued** function with $f = (f_1, \dots, f_m)$, we define the **Jacobian matrix** is the $m \times n$ matrix (sometimes denoted $J(x)$)

$$Df(\mathbf{x}) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix}$$

Examples:

- If $f(x, y) = x^2y + y^2x$,

$$Df(x, y) = (2xy + y^2, x^2 + 2yx)$$

- If $f(x, y) = (xy, x^2 + y^2)$,

$$Df(x, y) = \begin{pmatrix} y & x \\ 2x & 2y \end{pmatrix}$$

Differentiability in $\mathbb{R}^n$

Let $f : S \rightarrow T$, where $S \subset \mathbb{R}^n$, $T \subset \mathbb{R}^m$ a vector-valued function. As in the case of functions of one variable, f is differentiable at $\mathbf{x}$ if it can be approximated by a linear function around $\mathbf{x}$.

Differentiability

The function f is **differentiable** at $\mathbf{x}$ if there exists a $m \times n$ matrix A such that

$$\lim_{\|\mathbf{h}\| \rightarrow 0} \frac{\|f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) - A\mathbf{h}\|}{\|\mathbf{h}\|} = 0$$

- A is the **(total) derivative** of f at $\mathbf{x}$.
- The map $\mathbf{x} + \mathbf{h} \mapsto f(\mathbf{x}) + A\mathbf{h}$ is the best linear approximation to f at $\mathbf{x}$. Its graph is the **tangent plane** to f at $\mathbf{x}$.
- If f is differentiable at $\mathbf{x}$, then it is continuous at $\mathbf{x}$, all its partial derivatives exist and $A = Df(\mathbf{x})$, the Jacobian matrix of f .

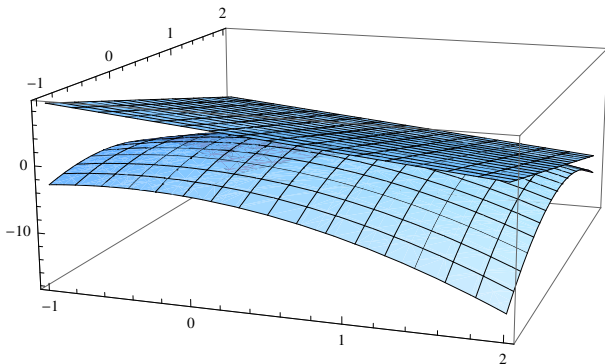
Differentiability in $\mathbb{R}^n$

Example:

Consider $f(x, y) = -2x^2 + 2xy - 3y^2$.

At $(1, 1)$, $f(1, 1) = -3$ and $Df(1, 1) = (-2, -4)$, so tangent plane is

$$z(x, y) = -3 - 2(x - 1) - 4(y - 1)$$



Differentiability in $\mathbb{R}^n$

Note that the existence of all the partial derivatives does not guarantee differentiability. In fact, not even continuity!

Example:

$$f(x, y) = \begin{cases} \frac{xy^2}{x^2+y^4} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Both partial derivatives exist at $(0, 0)$:

- $\lim_{h_1 \rightarrow 0} \frac{f(h_1, 0) - f(0, 0)}{h_1} = \lim_{h_1 \rightarrow 0} \frac{0 - 0}{h_1} = 0$
- $\lim_{h_2 \rightarrow 0} \frac{f(0, h_2) - f(0, 0)}{h_2} = \lim_{h_2 \rightarrow 0} \frac{0 - 0}{h_2} = 0.$

But f is not even continuous at 0! ($\{f(\frac{1}{n^2}, \frac{1}{n})\} \rightarrow \frac{1}{2}$ and $\{f(0, \frac{1}{n})\} \rightarrow 0$)

Differentiability in $\mathbb{R}^n$

Sufficient Condition for Differentiability

If all the partial derivatives of f **exist** in a neighbourhood of $\mathbf{x}$ and are **continuous** at $\mathbf{x}$, then f is differentiable at $\mathbf{x}$.

If all the partial derivatives **exist** and are **continuous** on S , f is said to be **continuously differentiable**, or C^1 , on S

$$C^1 \Rightarrow \text{Differentiable}$$

Differentiability in $\mathbb{R}^n$

A sketch proof: $C^1 \Rightarrow$ Differentiable

Consider a function of two variables $f(x, y)$.

$$\begin{aligned} f(x+h, y+k) - f(x, y) &= [f(x+h, y+k) - f(x, y+k)] + [f(x, y+k) - f(x, y)] \\ &= f_x(\eta, y+k)h + f_y(x, \xi)k \end{aligned}$$

where $\eta \in [x, x+h]$ and $\xi \in [y, y+k]$, applying the Mean Value Theorem. Hence

$$f(x+h, y+k) - f(x, y) = f_x(x, y)h + f_y(x, y)k + R(k, h)$$

where $R(k, h) = [f_x(\eta, y+k) - f_x(x, y)]h + [f_y(x, \xi) - f_y(x, y)]k$. As f_x and f_y are continuous and $\eta \rightarrow x$ and $\xi \rightarrow y$ as $k, h \rightarrow 0$,

$$R(k, h)/\sqrt{h^2 + k^2} \rightarrow_{k, h \rightarrow 0} 0$$

Hence f is differentiable at (x, y) .

Differentiability

Differentiable $\nRightarrow C^1$.

Example:

A function that is differentiable but not C^1 .

$$f(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

- f is differentiable everywhere, even at $x = 0$:

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} h \sin(1/h) = 0$$

- but f' is not continuous at 0.

$$f'(x) = \begin{cases} 2x \sin(1/x) - \cos(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Chain Rule

The chain rule for functions of one variable can be generalised to several dimensions.

Chain Rule

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$, and $\mathbf{x} : \mathbb{R}^m \rightarrow \mathbb{R}^n$ be C^1 functions. Define $z : \mathbb{R}^m \rightarrow \mathbb{R}$ such that $z(\mathbf{t}) = f(\mathbf{x}(\mathbf{t}))$, then z is C^1 , and

$$\frac{\partial z}{\partial t_k} = f_1 \frac{\partial x_1}{\partial t_k} + f_2 \frac{\partial x_2}{\partial t_k} + \cdots + f_n \frac{\partial x_n}{\partial t_k} \quad \text{for } 1 \leq k \leq m$$

Example: $f(x, y) = x^2 + y$ where $x(s, t) = s + t$ and $y(s, t) = s^2 + t^2$

- $\frac{\partial z}{\partial s} = 2x \cdot 1 + 1 \cdot 2s = 4s + 2t,$
- $\frac{\partial z}{\partial t} = 2x \cdot 1 + 1 \cdot 2t = 2s + 4t.$

Chain Rule

The chain rule can be expressed using differential notation:

Total Differential

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a C^1 function. If there is an infinitesimal change $d\mathbf{x}$ in $\mathbf{x}$, the corresponding change in f is given by the **total differential**:

$$df = f_1 dx_1 + f_2 dx_2 + \cdots + f_n dx_n \quad \text{or} \quad df = Df \cdot d\mathbf{x}$$

Example:

$$f(x, y) = x^2 y + y^2 x \quad \implies \quad df = (2xy + y^2)dx + (x^2 + 2yx)dy$$

Chain Rule

or... using matrix notation:

Chain rule

If $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at a and $g : \mathbb{R}^m \rightarrow \mathbb{R}^p$ is differentiable at $f(a)$ then $g \circ f : \mathbb{R}^n \rightarrow \mathbb{R}^p$ is differentiable at a and

$$D(g \circ f)(a) = Dg(f(a))Df(a).$$

Example: $f(s, t) = (s + t, s^2 + t^2)$, $g(x, y) = x^2 + y$;

$$z = g \circ f \text{ maps } (s, t) \mapsto g(f(s, t)) = g(s + t, s^2 + t^2).$$

$$Dz(s, t) = \left(\frac{\partial z}{\partial s} \quad \frac{\partial z}{\partial t} \right) = (2x \quad 1)|_{(s+t, s^2+t^2)} \begin{pmatrix} 1 & 1 \\ 2s & 2t \end{pmatrix} = (2(s+t) \quad 1) \begin{pmatrix} 1 & 1 \\ 2s & 2t \end{pmatrix} = (4s + 2t \quad 2s + 4t)$$

Directional Derivative

Using the chain rule we can derive the **directional derivative** of f in the direction $\mathbf{v}$. Given a C^1 function f ,

$$\lim_{t \rightarrow 0} \frac{f(\mathbf{x} + t\mathbf{v}) - f(\mathbf{x})}{t} = \frac{d}{dt} f(\mathbf{x} + t\mathbf{v})_{t=0} = \nabla f(\mathbf{x}) \cdot \mathbf{v}$$

Example: $f(x, y) = x^2 + 2xy$, $\mathbf{v} = (1, 2)$.

$$\frac{d}{dt} f(\mathbf{x} + t\mathbf{v})_{t=0} = \begin{pmatrix} 2x + 2y \\ 2x \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 2x + 2y + 4x = 6x + 2y$$

or in matrix notation: $\frac{d}{dt} f(\mathbf{x} + t\mathbf{v})_{t=0} = Df(\mathbf{x})\mathbf{v} = \begin{pmatrix} 2x + 2y & 2x \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 6x + 2y$

Higher Order Partial Derivatives

The partial derivatives of a function of several variables are themselves functions of several variables, so might be differentiated again.

For a real-valued function $f(\mathbf{x})$ of n variables:

- the first-order partial derivatives are: $\frac{\partial f}{\partial x_j}$ or $\partial_j f$ or $f_j(\mathbf{x})$, for $j = 1, \dots, n$;
- the second-order partial derivatives are: $\frac{\partial}{\partial x_i} \left(\frac{\partial f}{\partial x_j} \right) = \frac{\partial^2 f}{\partial x_i \partial x_j}$ or $\partial_i \partial_j f$ or $f_{ij}(\mathbf{x})$, for $i, j = 1, \dots, n$.
- k^{th} -order partial derivatives...

If all the partial derivatives of order less than or equal to k exist and are continuous on S , f is said to be k -times continuously differentiable, or C^k , on S

Second Order Partial Derivatives

Young's Theorem

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a C^2 function. Then

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i} \quad \text{for } i, j = 1, \dots, n$$

If the second partial derivatives are not continuous this need not be true:

$$f(x, y) = \begin{cases} xy \frac{x^2 - y^2}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

then one can check that f is C^1 but $\partial_y \partial_x f(0, 0) = -1$ whereas $\partial_x \partial_y f(0, 0) = 1$.

Hint: Show that $\frac{\partial f}{\partial x}(0, y) = -y$ and $\frac{\partial f}{\partial y}(x, 0) = x$

Hessian

The second-order partial derivatives can be written as a matrix, known as the **Hessian matrix**:

$$D^2f = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}$$

If f is a C^2 function the Hessian matrix is **symmetric**.

Example: $f(x, y) = x^2y - 3y^2x$

$$Df = (2xy - 3y^2, x^2 - 6xy) \text{ and } D^2f = \begin{pmatrix} 2y & 2x - 6y \\ 2x - 6y & -6x \end{pmatrix}$$

Taylor's Theorem in $\mathbb{R}^n$

Taylor's Theorem in $\mathbb{R}^n$

Suppose $f : U \rightarrow \mathbb{R}$, where U is an open subset of $\mathbb{R}^n$, is a C^2 function. Then for any $\mathbf{a}, \mathbf{x} \in U$:

$$f(\mathbf{x}) = f(\mathbf{a}) + Df(\mathbf{a})(\mathbf{x} - \mathbf{a}) + R_1(\mathbf{x}, \mathbf{a})$$

$$\text{and } f(\mathbf{x}) = f(\mathbf{a}) + Df(\mathbf{a})(\mathbf{x} - \mathbf{a}) + \frac{1}{2}(\mathbf{x} - \mathbf{a})^T D^2 f(\mathbf{a})(\mathbf{x} - \mathbf{a}) + R_2(\mathbf{x}, \mathbf{a})$$

where

$$\lim_{\mathbf{x} \rightarrow \mathbf{a}} \frac{R_1(\mathbf{x}, \mathbf{a})}{\|\mathbf{x} - \mathbf{a}\|} \rightarrow 0 \quad \text{and} \quad \lim_{\mathbf{x} \rightarrow \mathbf{a}} \frac{R_2(\mathbf{x}, \mathbf{a})}{\|\mathbf{x} - \mathbf{a}\|^2} \rightarrow 0$$

Taylor approximations in $\mathbb{R}^n$

Example:

Find the first and second-order approximations of $z = x^2y + y^2x$ at $(1, 2)$.

- First-order Taylor approximation:

$$z = 6 + 8(x - 1) + 5(y - 2) \implies z = 8x + 5y - 12$$

Since $f(1, 2) = 6$ and $Df(\mathbf{x}) = (2xy + y^2, x^2 + 2yx)$, so $Df(1, 2) = (8, 5)$.

- Second-order Taylor approximation:

$$f(x, y) = 6 + 8(x - 1) + 5(y - 2) + \frac{1}{2} [4(x - 1)^2 + 12(x - 1)(y - 2) + 2(y - 2)^2] + R_2$$

since the Hessian is $H(\mathbf{x}) = \begin{pmatrix} 2y & 2x + 2y \\ 2x + 2y & 2x \end{pmatrix}$, so $H(1, 2) = \begin{pmatrix} 4 & 6 \\ 6 & 2 \end{pmatrix}$.

III. Implicit Function Theorem

Comparative Statics

In many economic problems, the *solution*, or *equilibrium*, depends on one or more parameters, and we want to know how the solution changes when a parameter changes. This exercise is called **Comparative Statics**.

Sometimes we don't have an explicit functional form of the solution. We only know that it satisfies some equations. Still, we want to understand how the solutions changes with some parameters. In order to do that we use the **implicit differentiation**.

Implicit Functions

Consider the equation $F(x, y)=0$. For each value of x you could try to solve the equation, to obtain a corresponding value of y .

If this is possible (if there is a unique y for each x), we say that the equation defines y as an **implicit function** of x .

Examples:

$$F(x, y) = 3x^2 - 2y, \quad F(x, y) = y + \ln(y + 1) - x, \quad F(x, y) = x^2 + y^2 - 4$$

The derivative of an implicit function $y(x)$ defined by $F(x, y) = 0$ can be obtained by **implicit differentiation**: differentiate the whole equation wrt x , allowing for the fact that y is a function of x .

$$y + \ln(y + 1) - x = 0 \quad \Rightarrow \quad \frac{dy}{dx} + \frac{1}{y+1} \frac{dy}{dx} - 1 = 0$$

$$\Rightarrow \frac{dy}{dx} \left(\frac{y+2}{y+1} \right) = 1 \Rightarrow \frac{dy}{dx} = \frac{y+1}{y+2}$$

The Implicit Function Theorem

The Implicit Function Theorem

Suppose $\mathbf{x} \in \mathbb{R}^m$, and $(\mathbf{x}^*, y^*)$ is a solution to $F(\mathbf{x}, y) = 0$. Suppose that F is C^1 in an open ball around $(\mathbf{x}^*, y^*)$, with $\frac{\partial F}{\partial y}(\mathbf{x}^*, y^*) \neq 0$. Then there is a C^1 function $y = y(\mathbf{x})$ defined on an open ball around $\mathbf{x}^*$ such that:

- (i) $y(\mathbf{x}^*) = y^*$
- (ii) $F(\mathbf{x}, y(\mathbf{x})) = 0$
- (iii) $\frac{\partial y}{\partial x_i}(\mathbf{x}, y) = - \frac{\frac{\partial F}{\partial x_i}(\mathbf{x}, y)}{\frac{\partial F}{\partial y}(\mathbf{x}, y)}$

Implication: If $F(\mathbf{x}, y) = 0$ has a solution y^* for a particular $\mathbf{x}^*$, then it is legitimate to think of y as a function of $\mathbf{x}$ around that point, and you can determine how y changes from y^* when $\mathbf{x}$ changes a little from $\mathbf{x}^*$ by implicitly differentiating.

The Implicit Function Theorem

Example:

Consider the equation $x^2 + y^2 = 10$. It has a solution at $x = 3, y = -1$. Find $\frac{dy}{dx}$ at this point.

- Define $F(x, y) = x^2 + y^2 - 10$.
 - ▶ $\frac{\partial F}{\partial x}(x, y) = 2x$, continuous.
 - ▶ $\frac{\partial F}{\partial y}(x, y) = 2y$, continuous. So F is C^1 .
 - ▶ $\frac{\partial F}{\partial y}(3, -1) = -2 \neq 0$

Then there exists $y = y(x)$ such that $y(3) = -1$ and $\frac{dy}{dx} = -\frac{x}{y} = 3$.

- The implicit function theorem does not say that $y(x)$ is the only solution to the equation for a given value of x but rather that it is the only solution near $(3, -1)$
- At the point $y = 0, x = \sqrt{10}$, $F_y = 2y = 0$ and the conditions of the implicit function theorem are not met. If x is increased a little no solution exists. If x is decreased a little there are two equally good candidates for a solution.

The Implicit Function Theorem

Consider the system of n equations in $n + m$ variables:

$$\begin{aligned} F_1(x_1, x_2, \dots, x_m; y_1, y_2, \dots, y_n) &= 0 \\ F_2(x_1, x_2, \dots, x_m; y_1, y_2, \dots, y_n) &= 0 \\ &\vdots \\ F_n(x_1, x_2, \dots, x_m; y_1, y_2, \dots, y_n) &= 0 \end{aligned}$$

These equations implicitly define y_i as a function of $(x_1, \dots, x_m)$. Can we generalise the Implicit Function Theorem to see how y_k changes with x_j ?

The Implicit Function Theorem

Implicitly differentiating the i^{th} equation with respect to x_j :

$$\frac{\partial F_i}{\partial x_j} + \frac{\partial F_i}{\partial y_1} \frac{\partial y_1}{\partial x_j} + \frac{\partial F_i}{\partial y_2} \frac{\partial y_2}{\partial x_j} + \cdots + \frac{\partial F_i}{\partial y_n} \frac{\partial y_n}{\partial x_j} = 0$$

Therefore

$$\begin{pmatrix} \frac{\partial F_1}{\partial y_1} & \frac{\partial F_1}{\partial y_2} & \cdots & \frac{\partial F_1}{\partial y_n} \\ \frac{\partial F_2}{\partial y_1} & \frac{\partial F_2}{\partial y_2} & \cdots & \frac{\partial F_2}{\partial y_n} \\ \vdots & & \ddots & \vdots \\ \frac{\partial F_n}{\partial y_1} & \frac{\partial F_n}{\partial y_2} & \cdots & \frac{\partial F_n}{\partial y_n} \end{pmatrix} \begin{pmatrix} \frac{\partial y_1}{\partial x_j} \\ \frac{\partial y_2}{\partial x_j} \\ \vdots \\ \frac{\partial y_n}{\partial x_j} \end{pmatrix} = - \begin{pmatrix} \frac{\partial F_1}{\partial x_j} \\ \frac{\partial F_2}{\partial x_j} \\ \vdots \\ \frac{\partial F_n}{\partial x_j} \end{pmatrix}$$

and provided the Jacobian matrix is non-singular this system has a solution.

The Implicit Function Theorem

Implicit Function Theorem

Consider $\mathbf{F} : \mathbb{R}^{m+n} \rightarrow \mathbb{R}^n$. Suppose that $\mathbf{F}(\mathbf{x}^*, \mathbf{y}^*) = \mathbf{0}$ and that each F_i is C^1 around $(\mathbf{x}^*, \mathbf{y}^*)$, and $D_{\mathbf{y}}\mathbf{F}$ is invertible at this point. Then, there exists a C^1 function $\mathbf{y}(\mathbf{x})$ such that:

- (i) $\mathbf{y}(\mathbf{x}^*) = \mathbf{y}^*$
- (ii) $\mathbf{F}(\mathbf{x}, \mathbf{y}(\mathbf{x})) = \mathbf{0}$
- (iii) $D_{\mathbf{x}}\mathbf{y}(\mathbf{x}) = -[D_{\mathbf{y}}\mathbf{F}(\mathbf{x}, \mathbf{y}(\mathbf{x}))]^{-1} D_{\mathbf{x}}\mathbf{F}(\mathbf{x}, \mathbf{y}(\mathbf{x})).$

Summing Up

We have defined several concepts of multivariate calculus:

- Functions and Continuity
 - ▶ Weierstrass Theorem, Intermediate Value Theorem, Brouwer's Fixed Point Theorems
- Differentiation
 - ▶ Partial Derivatives, gradient, Jacobian Matrix
 - ▶ Directional derivatives,
 - ▶ C^1 Functions
 - ▶ Chain Rule
 - ▶ Second and higher Order Partial Derivatives, Hessian Matrix
 - ▶ Taylor Approximations.
- Implicit Differentiation.